

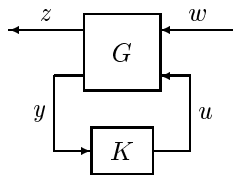
16

$\mathcal{H}_\infty$ Control: Simple Case

In this chapter we consider $\mathcal{H}_\infty$ control theory. Specifically, we formulate the optimal and suboptimal $\mathcal{H}_\infty$ control problems in section 16.1. However, we will focus on the suboptimal case in this book and discuss why we do so. In section 16.2 all suboptimal controllers are characterized for a class of simplified problems while leaving the more general problems to the next chapter. Some preliminary analysis is given in section 16.3 for the output feedback results. The analysis suggested the need for solving the Full Information and Output Estimation problems, which are the topics of sections 16.4-16.7. Section 16.8 discusses the $\mathcal{H}_\infty$ separation theory and presents the proof of the output feedback results. The behavior of the $\mathcal{H}_\infty$ controller as a function of performance level γ is considered in section 16.9. The optimal controllers are also briefly considered in this section. Some other interpretations of the $\mathcal{H}_\infty$ controllers are given in section 16.10. Finally, section 16.11 presents the formulas for an optimal $\mathcal{H}_\infty$ controller.

16.1 Problem Formulation

Consider the system described by the block diagram



where the plant G and controller K are assumed to be real-rational and proper. It will be assumed that state space models of G and K are available and that their realizations are assumed to be stabilizable and detectable. Recall again that a controller is said to be *admissible* if it internally stabilizes the system. Clearly, stability is the most basic requirement for a practical system to work. Hence any sensible controller has to be admissible.

Optimal $\mathcal{H}_\infty$ Control: *find all admissible controllers $K(s)$ such that $\|T_{zw}\|_\infty$ is minimized.*

It should be noted that the optimal $\mathcal{H}_\infty$ controllers as defined above are generally not unique for MIMO systems. Furthermore, finding an optimal $\mathcal{H}_\infty$ controller is often both numerically and theoretically complicated, as shown in Glover and Doyle [1989]. This is certainly in contrast with the standard $\mathcal{H}_2$ theory, in which the optimal controller is unique and can be obtained by solving two Riccati equations without iterations. Knowing the achievable optimal (minimum) $\mathcal{H}_\infty$ norm may be useful theoretically since it sets a limit on what we can achieve. However, in practice it is often not necessary and sometimes even undesirable to design an optimal controller, and it is usually much cheaper to obtain controllers that are very close in the norm sense to the optimal ones, which will be called *suboptimal controllers*. A suboptimal controller may also have other nice properties over optimal ones, e.g., lower bandwidth.

Suboptimal $\mathcal{H}_\infty$ Control: *Given $\gamma > 0$, find all admissible controllers $K(s)$ if there is any such that $\|T_{zw}\|_\infty < \gamma$.*

For the reasons mentioned above, we focus our attention in this book on suboptimal control. When appropriate, we briefly discuss what will happen when γ approaches the optimal value.

16.2 Output Feedback $\mathcal{H}_\infty$ Control

16.2.1 Internal Stability and Input/output Stability

Now suppose K is a stabilizing controller for the system G . Then the internal stability guarantees $T_{zw} = \mathcal{F}_\ell(G, K) \in \mathcal{RH}_\infty$, but the latter does not necessarily imply the internal stability. The following lemma provides the additional (mild) conditions to the equivalence of $T_{zw} = \mathcal{F}_\ell(G, K) \in \mathcal{RH}_\infty$ and internal stability of the closed-loop system. To state the lemma, we shall assume that G and K have the following stabilizable and detectable realizations, respectively:

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right], \quad K(s) = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array} \right].$$

Lemma 16.1 *Suppose that the realizations for G and K are both stabilizable and detectable. Then the feedback connection $T_{zw} = \mathcal{F}_\ell(G, K)$ of the realizations for G and K is*

(a) *detectable if $\begin{bmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all $\text{Re} \lambda \geq 0$;*

(b) *stabilizable if $\begin{bmatrix} A - \lambda I & B_1 \\ C_2 & D_{21} \end{bmatrix}$ has full row rank for all $\text{Re} \lambda \geq 0$.*

Moreover, if (a) and (b) hold, then K is an internally stabilizing controller iff $T_{zw} \in \mathcal{RH}_\infty$.

Proof. The state-space equations for the closed-loop are:

$$\begin{aligned} \mathcal{F}_\ell(G, K) &= \left[\begin{array}{cc|c} A + B_2 \hat{D} L_1 C_2 & B_2 L_2 \hat{C} & B_1 + B_2 \hat{D} L_1 D_{21} \\ \hat{B} L_1 C_2 & \hat{A} + \hat{B} L_1 D_{22} \hat{C} & \hat{B} L_1 D_{21} \\ \hline C_1 + D_{12} L_2 \hat{D} C_2 & D_{12} L_2 \hat{C} & D_{11} + D_{12} \hat{D} L_1 D_{21} \end{array} \right] \\ &=: \left[\begin{array}{c|c} A_c & B_c \\ \hline C_c & D_c \end{array} \right] \end{aligned}$$

where $L_1 := (I - D_{22} \hat{D})^{-1}$, $L_2 := (I - \hat{D} D_{22})^{-1}$.

Suppose $\mathcal{F}_\ell(G, K)$ has undetectable state $(x', y)'$ and mode $\text{Re} \lambda \geq 0$; then the PBH test gives

$$\begin{bmatrix} A_c - \lambda I \\ C_c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

This can be simplified as

$$\begin{bmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{bmatrix} \begin{bmatrix} x \\ \hat{D} L_1 C_2 x + L_2 \hat{C} y \end{bmatrix} = 0$$

and

$$\hat{B} L_1 (C_2 x + D_{22} \hat{C} y) + \hat{A} y - \lambda y = 0.$$

Now if

$$\begin{bmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{bmatrix}$$

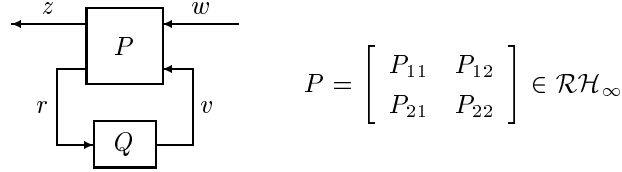
has full column rank, then $x = 0$ and $\hat{C} y = 0$. This implies $\hat{A} y = \lambda y$. Since $(\hat{C}, \hat{A})$ is detectable, we get $y = 0$, which is a contradiction. Hence part (a) is proven, and part (b) is a dual result. $\square$

These relations will be used extensively below to simplify our development and to enable us to focus on input/output stability only.

16.2.2 Contraction and Stability

One of the keys to the entire development of $\mathcal{H}_\infty$ theory is the fact that the contraction and internal stability is preserved under an inner linear fractional transformation.

Theorem 16.2 *Consider the following feedback system:*



Suppose that $P \sim P = I$, $P_{21}^{-1} \in \mathcal{RH}_\infty$ and that Q is a proper rational matrix. Then the following are equivalent:

- (a) The system is internally stable, well-posed, and $\|T_{zw}\|_\infty < 1$.
- (b) $Q \in \mathcal{RH}_\infty$ and $\|Q\|_\infty < 1$.

Proof. (b) $\Rightarrow$ (a). Note that since $P, Q \in \mathcal{RH}_\infty$, the system internal stability is guaranteed if $(I - P_{22}Q)^{-1} \in \mathcal{RH}_\infty$. Therefore, internal stability and well-posedness follow easily from $\|P_{22}\|_\infty \leq 1$, $\|Q\|_\infty < 1$, and a small gain argument. (Note that $\|P_{22}\|_\infty \leq 1$ follows from the fact that P_{22} is a compression of P .)

To show that $\|T_{zw}\|_\infty < 1$, consider the closed-loop system at any frequency $s = j\omega$ with the signals fixed as complex constant vectors. Let $\|Q\|_\infty =: \epsilon < 1$ and note that $T_{wr} = P_{21}^{-1}(I - P_{22}Q) \in \mathcal{RH}_\infty$. Also let $\kappa := \|T_{wr}\|_\infty$. Then $\|w\| \leq \kappa\|r\|$, and P inner implies that $\|z\|^2 + \|r\|^2 = \|w\|^2 + \|v\|^2$. Therefore,

$$\|z\|^2 \leq \|w\|^2 + (\epsilon^2 - 1)\|r\|^2 \leq [1 - (1 - \epsilon^2)\kappa^{-2}]\|w\|^2$$

which implies $\|T_{zw}\|_\infty < 1$.

(a) $\Rightarrow$ (b). To show that $\|Q\|_\infty < 1$, suppose there exist a (real or infinite) frequency ω and a constant nonzero vector r such that at $s = j\omega$, $\|Qr\| \geq \|r\|$. Then setting $w = P_{21}^{-1}(I - P_{22}Q)r$, $v = Qr$ gives $v = T_{vw}w$. But as above, P inner implies that $\|z\|^2 + \|r\|^2 = \|w\|^2 + \|v\|^2$ and, hence, that $\|z\|^2 \geq \|w\|^2$, which is impossible since $\|T_{zw}\|_\infty < 1$. It follows that $\sigma_{\max}(Q(j\omega)) < 1$ for all ω , i.e., $\|Q\|_\infty < 1$ since Q is rational.

To show $Q \in \mathcal{RH}_\infty$, let $Q = NM^{-1}$ with $N, M \in \mathcal{RH}_\infty$ be a right coprime factorization, i.e., there exist $X, Y \in \mathcal{RH}_\infty$ such that $XN + YM = I$. We shall show that $M^{-1} \in \mathcal{RH}_\infty$. By internal stability we have

$$Q(I - P_{22}Q)^{-1} = N(M - P_{22}N)^{-1} \in \mathcal{RH}_\infty$$

and

$$(I - P_{22}Q)^{-1} = M(M - P_{22}N)^{-1} \in \mathcal{RH}_\infty.$$

Thus

$$XQ(I - P_{22}Q)^{-1} + Y(I - P_{22}Q)^{-1} = (M - P_{22}N)^{-1} \in \mathcal{RH}_\infty.$$

This implies that the winding number of $\det(M - P_{22}N)$, as s traverses the Nyquist contour, equals zero. Now note the fact that, for all $s = j\omega$, $\det M^{-1} \neq 0$, $\det(I - \alpha P_{22}Q) \neq 0$ for all α in $[0,1]$ (this uses the fact that $\|P_{22}\|_\infty \leq 1$ and $\|Q\|_\infty < 1$). Also, $\det(I - \alpha P_{22}Q) = \det(M - \alpha P_{22}N) \det M^{-1}$, and we have $\det(M - \alpha P_{22}N) \neq 0$ for all α in $[0,1]$ and all $s = j\omega$. We conclude that the winding number of $\det M$ also equals zero. Therefore, $Q \in \mathcal{RH}_\infty$, and the proof is complete. $\square$

16.2.3 Simplifying Assumptions

In this chapter, we discuss a simplified version of $\mathcal{H}_\infty$ theory. The general case will be considered in the next chapter. The main reason for doing so is that the general case has its unique features but is much more involved algebraically. Involved algebra may distract attention from the essential ideas of the theory and therefore lose insight into the problem. Nevertheless, the problem considered below contains the essential features of the $\mathcal{H}_\infty$ theory.

The realization of the transfer matrix G is taken to be of the form

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ C_2 & D_{21} & 0 \end{array} \right].$$

The following assumptions are made:

- (i) (A, B_1) is stabilizable and (C_1, A) is detectable;
- (ii) (A, B_2) is stabilizable and (C_2, A) is detectable;
- (iii) $D_{12}^* \begin{bmatrix} C_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}$;
- (iv) $\begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^* = \begin{bmatrix} 0 \\ I \end{bmatrix}$.

Assumption (i) is made for a technical reason: together with (ii) it guarantees that the two Hamiltonian matrices (H_2 and J_2 in Chapter 14) in the $\mathcal{H}_2$ problem belong to $\text{dom}(\text{Ric})$. This assumption simplifies the theorem statements and proofs, but if it is relaxed, the theorems and proofs can be modified so that the given formulae are still correct, as will be seen in the next chapter. An important simplification that is a consequence of the assumption (i) is that internal stability is essentially equivalent to input-output stability ($T_{zw} \in \mathcal{RH}_\infty$). This equivalence enables us to focus only on input/output stability and is captured in Corollary 16.3 below. Of course, assumption (ii) is necessary and sufficient for G to be internally stabilizable, but is not needed

to prove the equivalence of internal stability and $T_{zw} \in \mathcal{RH}_\infty$. (Readers should be clear that this does not mean that the realization for G need not be stabilizable and detectable. In point of fact, the internal stability and input-output stability can never be equivalent if either G or K has unstabilizable or undetectable modes.)

Corollary 16.3 *Suppose that assumptions (i), (iii), and (iv) hold. Then a controller K is admissible iff $T_{zw} \in \mathcal{RH}_\infty$.*

Proof. The realization for plant G is stabilizable and detectable by assumption (i). We only need to verify that the rank conditions of the two matrices in Lemma 16.1 are satisfied. Now suppose assumptions (i) and (iii) are satisfied and let $D_\perp$ be such that $\begin{bmatrix} D_{12} & D_\perp \end{bmatrix}$ is a unitary matrix. Then

$$\begin{aligned} \text{rank} \begin{bmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{bmatrix} &= \text{rank} \begin{bmatrix} I & 0 \\ 0 & \begin{bmatrix} D_{12}^* \\ D_\perp^* \end{bmatrix} \end{bmatrix} \begin{bmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} A - \lambda I & B_2 \\ \begin{bmatrix} 0 \\ D_\perp^* C_1 \end{bmatrix} & \begin{bmatrix} I \\ 0 \end{bmatrix} \end{bmatrix}. \end{aligned}$$

So

$$\begin{bmatrix} A - \lambda I & B_2 \\ C_1 & D_{12} \end{bmatrix}$$

has full column rank for all $\text{Re} \lambda \geq 0$ iff

$$\begin{bmatrix} A - \lambda I \\ D_\perp^* C_1 \end{bmatrix}$$

has full column rank. However, the last matrix has full rank for all $\text{Re} \lambda \geq 0$ iff $(D_\perp^* C_1, A)$ is detectable. Since $D_\perp(D_\perp^* C_1) = (I - D_{12} D_{12}^*) C_1 = C_1$, $(D_\perp^* C_1, A)$ is detectable iff (C_1, A) is detectable. The rank condition for the other matrix follows by duality. $\square$

Two additional assumptions that are implicit in the assumed realization for $G(s)$ are that $D_{11} = 0$ and $D_{22} = 0$. As we have mentioned many times, $D_{22} \neq 0$ does not pose any problem since it is easy to form an equivalent problem with $D_{22} = 0$ by a linear fractional transformation on the controller $K(s)$. However, relaxing the assumption $D_{11} = 0$ complicates the formulae substantially, as will be seen in the next chapter.

16.2.4 Suboptimal $\mathcal{H}_\infty$ Controllers

In this subsection, we present the necessary and sufficient conditions for the existence of an admissible controller $K(s)$ such that $\|T_{zw}\|_\infty < \gamma$ for a given γ , and, furthermore,

if the necessary and sufficient conditions are satisfied, we characterize all admissible controllers that satisfy the norm condition. The proofs of these results will be given in the later sections. Let $\gamma_{opt} := \min \{\|T_{zw}\|_\infty : K(s) \text{ admissible}\}$, i.e., the optimal level. Then, clearly, γ must be greater than γ_{opt} for the existence of suboptimal $\mathcal{H}_\infty$ controllers. In Section 16.9 we will briefly discuss how to find an admissible K to minimize $\|T_{zw}\|_\infty$. Optimal $\mathcal{H}_\infty$ controllers are more difficult to characterize than suboptimal ones, and this is one major difference between the $\mathcal{H}_\infty$ and $\mathcal{H}_2$ results. Recall that similar differences arose in the norm computation problem as well.

The $\mathcal{H}_\infty$ solution involves the following two Hamiltonian matrices:

$$H_\infty := \begin{bmatrix} A & \gamma^{-2}B_1B_1^* - B_2B_2^* \\ -C_1^*C_1 & -A^* \end{bmatrix}, \quad J_\infty := \begin{bmatrix} A^* & \gamma^{-2}C_1^*C_1 - C_2^*C_2 \\ -B_1B_1^* & -A \end{bmatrix}.$$

The important difference here from the $\mathcal{H}_2$ problem is that the (1,2)-blocks are not sign definite, so we cannot use Theorem 13.7 in Chapter 13 to guarantee that $H_\infty \in \text{dom}(\text{Ric})$ or $\text{Ric}(H_\infty) \geq 0$. Indeed, these conditions are intimately related to the existence of $\mathcal{H}_\infty$ suboptimal controllers. Note that the (1,2)-blocks are a suggestive combination of expressions from the $\mathcal{H}_\infty$ norm characterization in Chapter 4 (or bounded real ARE in Chapter 13) and from the $\mathcal{H}_2$ synthesis of Chapter 14. It is also clear that if γ approaches infinity, then these two Hamiltonian matrices become the corresponding $\mathcal{H}_2$ control Hamiltonian matrices. The reasons for the form of these expressions should become clear through the discussions and proofs for the following theorem.

Theorem 16.4 *There exists an admissible controller such that $\|T_{zw}\|_\infty < \gamma$ iff the following three conditions hold:*

- (i) $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty := \text{Ric}(H_\infty) \geq 0$;
- (ii) $J_\infty \in \text{dom}(\text{Ric})$ and $Y_\infty := \text{Ric}(J_\infty) \geq 0$;
- (iii) $\rho(X_\infty Y_\infty) < \gamma^2$.

Moreover, when these conditions hold, one such controller is

$$K_{sub}(s) := \left[\begin{array}{c|c} \hat{A}_\infty & -Z_\infty L_\infty \\ \hline F_\infty & 0 \end{array} \right]$$

where

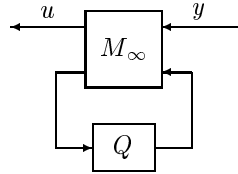
$$\begin{aligned} \hat{A}_\infty &:= A + \gamma^{-2}B_1B_1^*X_\infty + B_2F_\infty + Z_\infty L_\infty C_2 \\ F_\infty &:= -B_2^*X_\infty, \quad L_\infty := -Y_\infty C_2^*, \quad Z_\infty := (I - \gamma^{-2}Y_\infty X_\infty)^{-1}. \end{aligned}$$

The $\mathcal{H}_\infty$ controller displayed in Theorem 16.4, which is often called the *central controller* or *minimum entropy controller*, has certain obvious similarities to the $\mathcal{H}_2$ controller as well as some important differences. Although not as apparent as in the $\mathcal{H}_2$ case, the $\mathcal{H}_\infty$ controller also has an interesting separation structure. Furthermore,

each of the conditions in the theorem can be given a system-theoretic interpretation in terms of this separation. These interpretations, given in Section 16.8, require the filtering and full information (i.e., state feedback) results in sections 16.7 and 16.4. The proof of Theorem 16.4 is constructed out of these results as well. The term central controller will be obvious from the parameterization of all suboptimal controllers given below, while the meaning of minimum entropy will be discussed in Section 16.10.1.

The following theorem parameterizes the controllers that achieve a suboptimal $\mathcal{H}_\infty$ norm less than γ .

Theorem 16.5 *If conditions (i) to (iii) in Theorem 16.4 are satisfied, the set of all admissible controllers such that $\|T_{zw}\|_\infty < \gamma$ equals the set of all transfer matrices from y to u in*



$$M_\infty(s) = \left[\begin{array}{c|cc} \hat{A}_\infty & -Z_\infty L_\infty & Z_\infty B_2 \\ \hline F_\infty & 0 & I \\ -C_2 & I & 0 \end{array} \right]$$

where $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$.

As in the $\mathcal{H}_2$ case, the suboptimal controllers are parameterized by a fixed linear-fractional transformation with a free parameter Q . With $Q = 0$ (at the “center” of the set $\|Q\|_\infty < \gamma$), we recover the central controller $K_{sub}(s)$.

16.3 Motivation for Special Problems

Although the proof for output feedback results will be given later, we shall now try to give some ideas for approaching the problem. Specifically, we try to motivate the study of the OE (and hence other special problems) and show how this problem arises naturally in proving the output feedback results. The key is to use the fact that contraction and internal stability are preserved under an inner linear fractional transformation, which is Theorem 16.2. Assuming that X_∞ exists, we will show that the general output feedback problem boils down to an output estimation problem which can be solved easily if a state feedback or full information control problem can be solved.

Suppose $X_\infty := Ric(H_\infty)$ exists. Then X_∞ satisfies the following Riccati equation:

$$A^* X_\infty + X_\infty A + C_1^* C_1 + \gamma^{-2} X_\infty B_1 B_1^* X_\infty - X_\infty B_2 B_2^* X_\infty = 0. \quad (16.1)$$

Let x denote the state of G with respect to a given input w , and then we can differentiate $x(t)^* X_\infty x(t)$:

$$\frac{d}{dt}(x^* X_\infty x) = \dot{x}^* X_\infty x + x^* X_\infty \dot{x}$$

$$= x^*(A^*X_\infty + X_\infty A)x + 2\langle w, B_1^*X_\infty x \rangle + 2\langle u, B_2^*X_\infty x \rangle.$$

Using the Riccati equation for X_∞ to substitute in for $A^*X_\infty + X_\infty A$ gives

$$\frac{d}{dt}(x^*X_\infty x) = -\|C_1 x\|^2 - \gamma^{-2}\|B_1^*X_\infty x\|^2 + \|B_2^*X_\infty x\|^2 + 2\langle w, B_1^*X_\infty x \rangle + 2\langle u, B_2^*X_\infty x \rangle.$$

Finally, completion of the squares along with orthogonality of $C_1 x$ and $D_{12}u$ gives the key equation

$$\frac{d}{dt}(x^*X_\infty x) = -\|z\|^2 + \gamma^2\|w\|^2 - \gamma^2\|w - \gamma^{-2}B_1^*X_\infty x\|^2 + \|u + B_2^*X_\infty x\|^2. \quad (16.2)$$

Assume $x(0) = x(\infty) = 0$, $w \in \mathcal{L}_{2+}$, and integrate (16.2) from $t = 0$ to $t = \infty$:

$$\|z\|_2^2 - \gamma^2\|w\|_2^2 = \|u + B_2^*X_\infty x\|_2^2 - \gamma^2\|w - \gamma^{-2}B_1^*X_\infty x\|_2^2 = \|v\|_2^2 - \gamma^2\|r\|_2^2 \quad (16.3)$$

where

$$v := u + B_2^*X_\infty x, \quad r := w - \gamma^{-2}B_1^*X_\infty x. \quad (16.4)$$

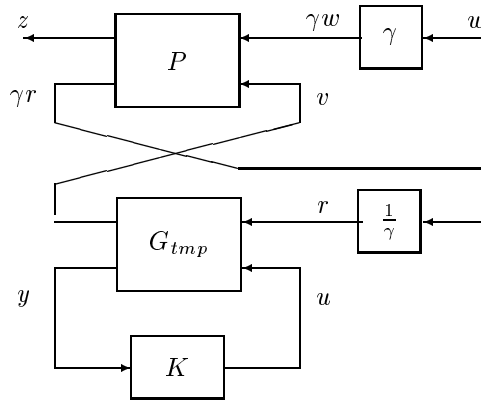
With these new defined variables, the closed-loop system can be expressed as two interconnected subsystems below:

$$\begin{bmatrix} \dot{x} \\ z \\ \gamma r \end{bmatrix} = \begin{bmatrix} A_{F_\infty} & \gamma^{-1}B_1 & B_2 \\ C_{1F_\infty} & 0 & D_{12} \\ -\gamma^{-1}B_1^*X_\infty & I & 0 \end{bmatrix} \begin{bmatrix} x \\ \gamma w \\ v \end{bmatrix} \quad \begin{array}{l} A_{F_\infty} := A + B_2F_\infty \\ C_{1F_\infty} := C_1 + D_{12}F_\infty \end{array}$$

and

$$\begin{bmatrix} \dot{x} \\ v \\ y \end{bmatrix} = \begin{bmatrix} A_{tmp} & B_1 & B_2 \\ -F_\infty & 0 & I \\ C_2 & D_{21} & 0 \end{bmatrix} \begin{bmatrix} x \\ r \\ u \end{bmatrix} \quad A_{tmp} := A + \gamma^{-2}B_1B_1^*X_\infty$$

where F_∞ is defined as in Section 16.2.4. This is shown in the following diagram:



where

$$P := \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{c|cc} A_{F_\infty} & \gamma^{-1}B_1 & B_2 \\ \hline C_{1F_\infty} & 0 & D_{12} \\ -\gamma^{-1}B_1^*X_\infty & I & 0 \end{array} \right] \quad (16.5)$$

and

$$G_{tmp} = \left[\begin{array}{c|cc} A_{tmp} & B_1 & B_2 \\ \hline -F_\infty & 0 & I \\ C_2 & D_{21} & 0 \end{array} \right]. \quad (16.6)$$

The equality (16.3) motivates the change of variables to r and v as in (16.4), and these variables provide the connection between T_{zw} and T_{vr} . Note that $T_{z(\gamma w)} = \gamma^{-1}T_{zw}$ and $T_{v(\gamma r)} = \gamma^{-1}T_{vr}$. It is immediate from equality (16.3) that $\|T_{zw}\|_\infty \leq \gamma$ iff $\|T_{vr}\|_\infty \leq \gamma$. While this is the basic idea behind the proof of Lemma 16.8 below, the details needed for strict inequality and internal stability require a bit more work.

Note that $w_{worst} := \gamma^{-2}B_1^*X_\infty x$ is the worst disturbance input in the sense that it maximizes the quantity $\|z\|_2^2 - \gamma^2\|w\|_2^2$ in (16.3) for the minimizing value of $u = -B_2^*X_\infty x$; that is, the u making $v = 0$ and the w making $r = 0$ are values satisfying a saddle point condition. (See Section 17.8 for more interpretations.) It is also interesting to note that w_{worst} is the optimal strategy for w in the corresponding LQ game problem (see the differential game problem in the last chapter). Equation (16.3) also suggests that $u = -B_2^*X_\infty x$ is a suboptimal control for a full information (FI) problem if the state x is available. This will be shown later. In terms of the OE problem for G_{tmp} , the output being estimated is the optimal FI control input $F_\infty x$ and the new disturbance r is offset by the “worst case” FI disturbance input w_{worst} .

Notice the structure of G_{tmp} : it is an OE problem. We will show below that the output feedback can indeed be transformed into the OE problem. To show this we first need to prove some preliminary facts.

Lemma 16.6 *Suppose $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty)$. Then $A_{F_\infty} = A + B_2F_\infty$ is stable iff $X_\infty \geq 0$.*

Proof. Re-arrange the Riccati equation for X_∞ and use the definition of F_∞ and C_{1F_∞} to get

$$A_{F_\infty}^*X_\infty + X_\infty A_{F_\infty} + \begin{bmatrix} C_{1F_\infty} \\ -\gamma^{-1}B_1^*X_\infty \end{bmatrix}^* \begin{bmatrix} C_{1F_\infty} \\ -\gamma^{-1}B_1^*X_\infty \end{bmatrix} = 0. \quad (16.7)$$

Since $H_\infty \in \text{dom}(\text{Ric})$, $(A_{F_\infty} + \gamma^{-2}B_1B_1^*X_\infty)$ is stable and hence $(B_1^*X_\infty, A_{F_\infty})$ is detectable. Then from standard results on Lyapunov equations (see Lemma 3.19), A_{F_∞} is stable iff $X_\infty \geq 0$. $\square$

Equation (16.3) can be written as

$$\|z\|_2^2 + \|\gamma r\|_2^2 = \|\gamma w\|_2^2 + \|v\|_2^2.$$

This suggests that P might be inner when $X_\infty \geq 0$, which is verified by the following lemma.

Lemma 16.7 *If $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$, then P in (16.5) is in $\mathcal{RH}_\infty$ and inner, and $P_{21}^{-1} \in \mathcal{RH}_\infty$.*

Proof. By Lemma 16.6, A_{F_∞} is stable. So $P \in \mathcal{RH}_\infty$. That P is inner ($P^*P = I$) follows from Lemma 13.29 upon noting that the observability Gramian of P is X_∞ (see (16.7)) and

$$\begin{bmatrix} 0 & I \\ D_{12}^* & 0 \end{bmatrix} \begin{bmatrix} C_{1F_\infty} \\ -\gamma^{-1}B_1^*X_\infty \end{bmatrix} + \begin{bmatrix} \gamma^{-1}B_1^* \\ B_2^* \end{bmatrix} X_\infty = 0.$$

Finally, the state matrix for P_{21}^{-1} is $(A_{F_\infty} + \gamma^{-2}B_1B_1^*X_\infty)$, which is stable by definition. Thus, $P_{21}^{-1} \in \mathcal{RH}_\infty$. $\square$

The following lemma connects these two systems T_{zw} and T_{vr} , which is the central part of the separation argument in Section 16.8.2. Recall that internal and input-output stability are equivalent for admissibility of K in the output feedback problem by Corollary 16.3.

Lemma 16.8 *Assume $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$. Then K is admissible for G and $\|T_{zw}\|_\infty < \gamma$ iff K is admissible for G_{tmp} and $\|T_{vr}\|_\infty < \gamma$.*

Proof. We may assume without loss of generality that the realization of K is stabilizable and detectable. Recall from Corollary 16.3 that internal stability for T_{zw} is equivalent to $T_{zw} \in \mathcal{RH}_\infty$. Similarly since

$$\begin{bmatrix} A_{tmp} - \lambda I & B_1 \\ C_2 & D_{21} \end{bmatrix} = \begin{bmatrix} A - \lambda I & B_1 \\ C_2 & D_{21} \end{bmatrix} \begin{bmatrix} I & 0 \\ \gamma^{-2}B_1^*X_\infty & I \end{bmatrix}$$

has full row rank for all $\text{Re}(\lambda) \geq 0$ and since

$$\det \begin{bmatrix} A_{tmp} - \lambda I & B_2 \\ -F_\infty & I \end{bmatrix} = \det(A_{tmp} + B_2F_\infty - \lambda I) \neq 0$$

for all $\text{Re}(\lambda) \geq 0$ by the stability of $A_{tmp} + B_2F_\infty$, we have that

$$\begin{bmatrix} A_{tmp} - \lambda I & B_2 \\ -F_\infty & I \end{bmatrix}$$

has full column rank. Hence by Lemma 16.1 the internal stability of T_{vr} , i.e., the internal stability of the subsystem consisting of G_{tmp} and controller K , is also equivalent to $T_{vr} \in \mathcal{RH}_\infty$. Thus internal stability is equivalent to input-output stability for both

G and G_{tmp} . This shows that K is an admissible controller for G if and only if it is admissible for G_{tmp} . Now it follows from Theorem 16.2 and Lemma 16.7 along with the above block diagram that $\|T_{z(\gamma w)}\|_\infty < 1$ iff $\|T_{v(\gamma r)}\|_\infty < 1$ or, equivalently, $\|T_{zw}\|_\infty < \gamma$ iff $\|T_{vr}\|_\infty < \gamma$. $\square$

From the previous analysis, it is clear that to solve the output feedback problem we need to show

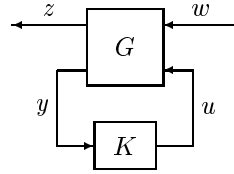
(a) $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$;

(b) $\|T_{vr}\|_\infty < \gamma$.

To show (a), we need to solve a FI problem. The problem (b) is an OE problem which can be solved by using the relationship between FC and OE problems in Chapter 12, while the FC problem can be solved by using the FI solution through duality. So in the sections to follow, we will focus on these special problems.

16.4 Full Information Control

Our system diagram in this section is standard as before



with

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ \left[\begin{array}{c} I \\ 0 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] & \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \end{array} \right].$$

The $\mathcal{H}_\infty$ problem corresponding to this setup again is not, strictly speaking, a special case of the output feedback problem because it does not satisfy all of the assumptions. In particular, it should be noted that for the FI (and FC in the next section) problem, internal stability is not equivalent to $T_{zw} \in \mathcal{RH}_\infty$ since

$$\left[\begin{array}{c|c} A - \lambda I & B_1 \\ \hline \left[\begin{array}{c} I \\ 0 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] \end{array} \right]$$

can never have full row rank, although this presents no difficulties in solving this problem. We simply must remember that in the FI case, K admissible means internally stabilizing, not just $T_{zw} \in \mathcal{RH}_\infty$.

We have seen that in the $\mathcal{H}_2$ FI case, the optimal controller uses just the state x even though the controller is provided with full information. We will show below that, in the $\mathcal{H}_\infty$ case, a suboptimal controller exists which also uses just x . This case could have been restricted to state feedback, which is more traditional, but we believe that, once one gets outside the pure $\mathcal{H}_2$ setting, the full information problem is more fundamental and more natural than the state feedback problem.

One setting in which the full information case is more natural occurs when the parameterization of all suboptimal controllers is considered. It is also appropriate when studying the general case when $D_{11} \neq 0$ in the next chapter or when $\mathcal{H}_\infty$ optimal (not just suboptimal) controllers are desired. Even though the optimal problem is not studied in detail in this book, we want the methods to extend to the optimal case in a natural and straightforward way.

The assumptions relevant to the FI problem which are inherited from the output feedback problem are

- (i) (C_1, A) is detectable;
- (ii) (A, B_2) is stabilizable;
- (iii) $D_{12}^* \begin{bmatrix} C_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}$.

Assumptions (iv) and the second part of (ii) for the general output feedback case have been effectively strengthened because of the assumed structure for C_2 and D_{21} .

Theorem 16.9 *There exists an admissible controller $K(s)$ for the FI problem such that $\|T_{zw}\|_\infty < \gamma$ if and only if $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$. Furthermore, if these conditions are satisfied, then the equivalence class of all admissible controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as*

$$K(s) \cong \begin{bmatrix} F_\infty - \gamma^{-2}Q(s)B_1^*X_\infty & Q(s) \end{bmatrix} \quad (16.8)$$

where $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$.

It is easy to see by comparing the $\mathcal{H}_\infty$ solution with the corresponding $\mathcal{H}_2$ solution that a fundamental difference between $\mathcal{H}_2$ and $\mathcal{H}_\infty$ controllers is that the $\mathcal{H}_\infty$ controller depends on the disturbance through B_1 whereas the $\mathcal{H}_2$ controller does not. This difference is essentially captured by the necessary and sufficient conditions for the existence of a controller given in Theorem 16.9. Note that these conditions are the same as condition (i) in Theorem 16.4.

The two individual conditions in Theorem 16.9 may each be given their own interpretations. The condition that $H_\infty \in \text{dom}(\text{Ric})$ implies that $X_\infty := \text{Ric}(H_\infty)$ exists

and $K(s) = \begin{bmatrix} F_\infty & 0 \end{bmatrix}$ gives T_{zw} as

$$T_{zw} = \left[\begin{array}{c|c} A_{F_\infty} & B_1 \\ \hline C_{1F_\infty} & 0 \end{array} \right] \quad \begin{array}{l} A_{F_\infty} = A + B_2 F_\infty \\ C_{1F_\infty} = C_1 + D_{12} F_\infty. \end{array} \quad (16.9)$$

Furthermore, since $T_{zw} = \gamma P_{11}$ and $P_{11}^\sim P_{11} = I - P_{21}^\sim P_{21}$ by $P^\sim P = I$, we have $\|P_{11}\|_\infty < 1$ and $\|T_{zw}\|_\infty = \gamma \|P_{11}\|_\infty < \gamma$. The further condition that $X_\infty \geq 0$ is equivalent, by Lemma 16.6, to this K stabilizing T_{zw} .

Remark 16.1 Given that $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$, there is an intuitive way to see why all the equivalence classes of controllers can be parameterized in the form of (16.8). Recall the following equality from equation (16.3):

$$\|z\|_2^2 - \gamma^2 \|w\|_2^2 = \|u + B_2^* X_\infty x\|_2^2 - \gamma^2 \|w - \gamma^{-2} B_1^* X_\infty x\|_2^2 = \|v\|_2^2 - \gamma^2 \|r\|_2^2.$$

$\|T_{zw}\|_\infty < \gamma$ means that $\|z\|_2^2 - \gamma^2 \|w\|_2^2 < 0$ for $w \neq 0$. This implies that $r \neq 0$ and $\|v\|_2^2 - \gamma^2 \|r\|_2^2 < 0$. Now all v satisfying this inequality can be written as $v = Qr$ for $Q \in \mathcal{RH}_\infty$ and $\|Q\|_\infty < \gamma$ or as $u + B_2^* X_\infty x = Q(w - \gamma^{-2} B_1^* X_\infty x)$. This gives equation (16.8). $\heartsuit$

We shall now prove the theorem.

Proof. ($\Rightarrow$) For simplicity, in the proof to follow we assume that the system is normalized such that $\gamma = 1$. Further, we will show that we can, without loss of generality, strengthen the assumption on (C_1, A) from detectable to observable. Suppose there exists an admissible controller $\hat{K} = \left[\begin{array}{c|cc} \hat{A} & \hat{B}_1 & \hat{B}_2 \\ \hline \hat{C} & \hat{D}_1 & \hat{D}_2 \end{array} \right]$ such that $\|T_{zw}\|_\infty < 1$. If (C_1, A) is

detectable but not observable, then change coordinates for the state of G to $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ with x_2 unobservable, (C_{11}, A_{11}) observable, and A_{22} stable, giving the following closed-loop state equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\hat{x}} \\ z \\ u \end{bmatrix} = \begin{bmatrix} A_{11} & 0 & 0 & B_{11} & B_{21} \\ A_{21} & A_{22} & 0 & B_{12} & B_{22} \\ \hat{B}_{11} & \hat{B}_{12} & \hat{A} & \hat{B}_2 & 0 \\ C_{11} & 0 & 0 & 0 & D_{12} \\ \hat{D}_{11} & \hat{D}_{12} & \hat{C} & \hat{D}_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \hat{x} \\ w \\ u \end{bmatrix}.$$

If we take a new plant G_{obs} with state x_1 and output z and group the rest of the equations as a new controller K_{obs} with the state made up of x_2 and $\hat{x}$, then

$$G_{obs}(s) = \left[\begin{array}{c|cc} A_{11} & B_{11} & B_{21} \\ \hline C_{11} & 0 & D_{12} \\ \hline \left[\begin{array}{c} I \\ 0 \end{array} \right] & \left[\begin{array}{c} 0 \\ I \end{array} \right] & \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \end{array} \right]$$

still satisfies the assumptions of the FI problem and is stabilized by K_{obs} with the closed-loop $\mathcal{H}_\infty$ norm $\|T_{zw}\|_\infty < 1$ where

$$K_{obs} = \left[\begin{array}{cc|cc} A_{22} + B_{22}\hat{D}_{12} & B_{22}\hat{C} & A_{21} + B_{22}\hat{D}_{11} & B_{12} + B_{22}\hat{D}_2 \\ \hline \hat{B}_{12} & \hat{A} & \hat{B}_{11} & \hat{B}_2 \\ \hline \hat{D}_{12} & \hat{C} & \hat{D}_{11} & \hat{D}_2 \end{array} \right].$$

If we now show that there exists $\hat{X}_\infty > 0$ solving the H_∞ Riccati equation for G_{obs} , i.e.,

$$\hat{X}_\infty = Ric \left[\begin{array}{cc} A_{11} & B_{11}B_{11}^* - B_{21}B_{21}^* \\ -C_{11}^*C_{11} & -A_{11}^* \end{array} \right],$$

then

$$Ric(H_\infty) = X_\infty = \left[\begin{array}{cc} \hat{X}_\infty & 0 \\ 0 & 0 \end{array} \right] \geq 0$$

exists for G . We can therefore assume without loss of generality that (C_1, A) is observable.

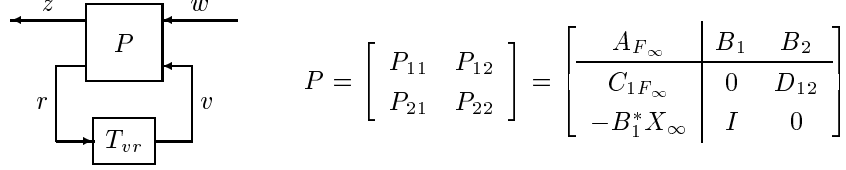
We shall suppose that there exists an admissible controller such that $\|T_{zw}\|_\infty < 1$. But note that the existence of an admissible controller such that $\|T_{zw}\|_\infty < 1$ is equivalent to that the admissible controller makes $\sup_{w \in \mathcal{BL}_{2+}} \|z\|_2 < 1$; hence, it is necessary that

$$\sup_{w \in \mathcal{BL}_{2+}} \min_{u \in \mathcal{L}_{2+}} \|z\|_2 < 1$$

since the latter is always no greater than the former by the fact that the set of signals u generated from admissible controllers is a subset of $\mathcal{L}_{2+}$. But from Theorem 15.9, the latter is true if and only if $H_\infty \in \text{dom}(Ric)$ and $X_\infty = Ric(H_\infty) > 0$. Hence the necessity is proven.

($\Leftarrow$) Suppose $H_\infty \in \text{dom}(Ric)$ and $X_\infty = Ric(H_\infty) \geq 0$ and suppose $K(s)$ is an admissible controller such that $\|T_{zw}\|_\infty < 1$. Again change variables to $v := u - F_\infty x$

and $r := w - B_1^* X_\infty x$, so that the closed-loop system is as shown below:



By Lemma 16.7, P is inner and $P_{21}^{-1} \in \mathcal{RH}_\infty$. By Theorem 16.2 the system is internally stable and $\|T_{zw}\|_\infty < 1$ iff $T_{vr} \in \mathcal{RH}_\infty$ and $\|T_{vr}\|_\infty < 1$. Now denote $Q := T_{vr}$, then $v = Qr$ and

$$\begin{aligned} v &= u - F_\infty x = \left(K - \begin{bmatrix} F_\infty & 0 \end{bmatrix} \right) \begin{bmatrix} x \\ w \end{bmatrix} \\ r &= w - B_1^* X_\infty x = \begin{bmatrix} -B_1^* X_\infty & I \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix}. \end{aligned}$$

Hence we have

$$\left(K - \begin{bmatrix} F_\infty & 0 \end{bmatrix} \right) \begin{bmatrix} x \\ w \end{bmatrix} = Q \begin{bmatrix} -B_1^* X_\infty & I \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix}$$

or

$$K - \begin{bmatrix} F_\infty & 0 \end{bmatrix} \cong Q \begin{bmatrix} -B_1^* X_\infty & I \end{bmatrix}.$$

Thus

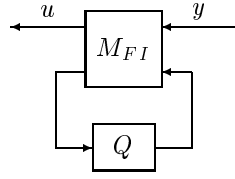
$$K(s) \cong \begin{bmatrix} F_\infty - Q(s)B_1^* X_\infty & Q(s) \end{bmatrix}, \quad Q \in \mathcal{RH}_\infty, \|Q\|_\infty < 1.$$

Hence all suboptimal controllers are in the stated equivalence class. $\square$

Remark 16.2 It should be emphasized that the set of controllers given above does not parameterize all controllers although it is sufficient for the purpose of deriving the output feedback results, and that is why “equivalence class” is used. It is clear that there is a suboptimal controller $K_1 = \begin{bmatrix} F_1 & 0 \end{bmatrix}$ with $F_1 \neq F_\infty$; however, there is no choice of Q such that K_1 belongs to the set. Nevertheless, this problem will not occur in output feedback case. $\heartsuit$

The following theorem gives all full information controllers.

Theorem 16.10 *Suppose the condition in Theorem 16.9 is satisfied; then all admissible controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as $K = \mathcal{F}_\ell(M_{FI}, Q)$:*



$$M_{FI}(s) = \left[\begin{array}{c|cc} A + B_2 F_\infty & \begin{bmatrix} 0 & B_1 \end{bmatrix} & B_2 \\ \hline 0 & \begin{bmatrix} F_\infty & 0 \end{bmatrix} & I \\ \begin{bmatrix} -I \\ 0 \end{bmatrix} & \begin{bmatrix} I & 0 \\ -\gamma^{-2} B_1^* X_\infty & I \end{bmatrix} & 0 \end{array} \right]$$

where $Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \in \mathcal{RH}_\infty$ and $\|Q_2\|_\infty < \gamma$.

Remark 16.3 It is simple to verify that for $Q_1 = 0$, we have

$$K = \begin{bmatrix} F_\infty - \gamma^{-2} Q_2 B_1^* X_\infty & Q_2 \end{bmatrix},$$

which is the parameterization given in Theorem 16.9. The parameterization of all suboptimal FC controllers follows by duality and, therefore, are omitted. $\heartsuit$

Proof. We only need to show that $\mathcal{F}_\ell(M_{FI}, Q)$ with $\|Q_2\|_\infty < \gamma$ parameterizes all FI $\mathcal{H}_\infty$ suboptimal controllers. To show that, we shall make a change of variables as before:

$$v = u + B_2^* X_\infty x, \quad r = w - \gamma^{-2} B_1^* X_\infty x.$$

Then the system equations can be written as follows:

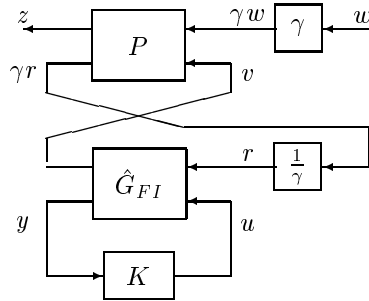
$$\begin{bmatrix} \dot{x} \\ z \\ \gamma r \end{bmatrix} = \begin{bmatrix} A_{F_\infty} & \gamma^{-1} B_1 & B_2 \\ C_{1F_\infty} & 0 & D_{12} \\ -\gamma^{-1} B_1^* X_\infty & I & 0 \end{bmatrix} \begin{bmatrix} x \\ \gamma w \\ v \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ v \\ y \end{bmatrix} = \begin{bmatrix} A_{tmp} & B_1 & B_2 \\ -F_\infty & 0 & I \\ \begin{bmatrix} I \\ \gamma^{-2} B_1^* X_\infty \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} & 0 \end{bmatrix} \begin{bmatrix} x \\ r \\ u \end{bmatrix}$$

where $A_{tmp} := A + \gamma^{-2} B_1 B_1^* X_\infty$.

This is shown pictorially in the following diagram:



where P is as given in equation (16.5) and

$$\hat{G}_{FI} = \left[\begin{array}{c|cc} A_{tmp} & B_1 & B_2 \\ \hline -F_\infty & 0 & I \\ I & \left[\begin{array}{c} 0 \\ I \end{array} \right] & 0 \\ \gamma^{-2}B_1^*X_\infty & \left[\begin{array}{c} 0 \\ I \end{array} \right] & 0 \end{array} \right].$$

So from Theorem 16.2 and Lemma 16.8, we conclude that K is an admissible controller for G and $\|T_{zw}\|_\infty < \gamma$ iff K is an admissible controller for $\hat{G}_{FI}$ and $\|T_{vr}\|_\infty < \gamma$.

Now let $L = \begin{bmatrix} B_2F_\infty & -B_1 \end{bmatrix}$; then $A_{tmp} + L \begin{bmatrix} I \\ \gamma^{-2}B_1^*X_\infty \end{bmatrix} = A + B_2F_\infty$ is stable.

Also note that $A_{tmp} + B_2F_\infty$ is stable. Then all controllers that stabilize $\hat{G}_{FI}$ can be parameterized as $K = \mathcal{F}_\ell(M, \Phi)$, $\Phi = \begin{bmatrix} \Phi_1 & \Phi_2 \end{bmatrix} \in \mathcal{RH}_\infty$ where

$$M = \left[\begin{array}{c|cc} A_{tmp} + B_2F_\infty + L \begin{bmatrix} I \\ \gamma^{-2}B_1^*X_\infty \end{bmatrix} & -L & B_2 \\ \hline F_\infty & 0 & I \\ - \begin{bmatrix} I \\ \gamma^{-2}B_1^*X_\infty \end{bmatrix} & I & 0 \end{array} \right].$$

With this parameterization of all controllers, the transfer matrix from r to v is $T_{vr} = \mathcal{F}_\ell(\hat{G}_{FI}, \mathcal{F}_\ell(M, \Phi)) =: \mathcal{F}_\ell(N, \Phi)$. It is easy to show that

$$N = \left[\begin{array}{c|c} 0 & I \\ \left[\begin{array}{c} 0 \\ I \end{array} \right] & 0 \end{array} \right]$$

and $T_{vr} = \mathcal{F}_\ell(N, \Phi) = \Phi_2$. Hence $\|T_{vr}\|_\infty < \gamma$ if and only if $\|\Phi_2\|_\infty < \gamma$. This implies that all FI $\mathcal{H}_\infty$ controllers can be parameterized as $K = \mathcal{F}_\ell(M, \Phi)$ with $\Phi = \begin{bmatrix} \Phi_1 & \Phi_2 \end{bmatrix} \in \mathcal{RH}_\infty$, $\|\Phi_2\|_\infty < \gamma$ and

$$M = \left[\begin{array}{c|cc} A + 2B_2F_\infty & - \begin{bmatrix} B_2F_\infty & -B_1 \end{bmatrix} & B_2 \\ \hline F_\infty & 0 & I \\ - \begin{bmatrix} I \\ \gamma^{-2}B_1^*X_\infty \end{bmatrix} & I & 0 \end{array} \right].$$

Now let

$$\Phi_1 = F_\infty - \gamma^{-2}Q_2B_1^*X_\infty + Q_1, \quad \Phi_2 = Q_2.$$

Then it is easy to show that $\mathcal{F}_\ell(M, \Phi) = \mathcal{F}_\ell(M_{FI}, Q)$. □

16.5 Full Control

$$G(s) = \left[\begin{array}{c|c|c} A & B_1 & \begin{bmatrix} I & 0 \end{bmatrix} \\ \hline C_1 & 0 & \begin{bmatrix} 0 & I \end{bmatrix} \\ \hline C_2 & D_{21} & \begin{bmatrix} 0 & 0 \end{bmatrix} \end{array} \right]$$

This problem is dual to the Full Information problem. The assumptions that the FC problem inherits from the output feedback problem are just the dual of those in the FI problem:

- (i) (A, B_1) is stabilizable;
- (ii) (C_2, A) is detectable;
- (iv) $\begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^* = \begin{bmatrix} 0 \\ I \end{bmatrix}$.

Theorem 16.11 *There exists an admissible controller $K(s)$ for the FC problem such that $\|T_{zw}\|_\infty < \gamma$ if and only if $J_\infty \in \text{dom}(\text{Ric})$ and $Y_\infty = \text{Ric}(J_\infty) \geq 0$. Moreover, if these conditions are satisfied then an equivalence class of all admissible controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as*

$$K(s) \cong \begin{bmatrix} L_\infty - \gamma^{-2} Y_\infty C_1^* Q(s) \\ Q(s) \end{bmatrix}$$

where $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$.

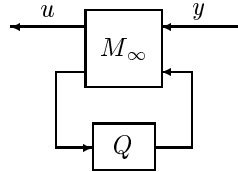
As expected, the condition in Theorem 16.11 is the same as that in (ii) of Theorem 16.4.

16.6 Disturbance Feedforward

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ \hline C_2 & I & 0 \end{array} \right]$$

This problem inherits the same assumptions (i)-(iii) as in the FI problem, but for internal stability we shall add that $A - B_1 C_2$ is stable. With this assumption, it is easy to check that the condition in Lemma 16.1 is satisfied so that the internal stability is again equivalent to $T_{zw} \in \mathcal{RH}_\infty$, as in the output feedback case.

Theorem 16.12 *There exists an admissible controller for the DF problem such that $\|T_{zw}\|_\infty < \gamma$ if and only if $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$. Moreover, if these conditions are satisfied then all admissible controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as the set of all transfer matrices from y to u in*



$$M_\infty(s) = \left[\begin{array}{c|cc} A + B_2 F_\infty - B_1 C_2 & B_1 & B_2 \\ \hline F_\infty & 0 & I \\ -C_2 - \gamma^{-2} B_1^* X_\infty & I & 0 \end{array} \right]$$

with $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$

Proof. Suppose there is a controller K_{DF} solving the above problem, i.e., with $\|T_{zw}\|_\infty < \gamma$. Then by Theorem 12.4, the controller $K_{FI} = K_{DF} \begin{bmatrix} C_2 & I \end{bmatrix}$ solves the corresponding $\mathcal{H}_\infty$ FI problem. Hence the conditions $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$ are necessarily satisfied. On the other hand, if these conditions are satisfied then FI is solvable. It is easy to verify that $\mathcal{F}_\ell(M_\infty, Q) = \mathcal{F}_\ell(P_{DF}, K_{FI})$ with $K_{FI} = \begin{bmatrix} F_\infty - \gamma^{-2} Q(s) B_1^* X_\infty & Q(s) \end{bmatrix}$ where P_{DF} is as defined in section 12.2 of Chapter 12. So again by Theorem 12.4, the controller $\mathcal{F}_\ell(M_\infty, Q)$ solves the DF problem.

To show that $\mathcal{F}_\ell(M_\infty, Q)$ with $\|Q\|_\infty < \gamma$ parameterizes all DF $\mathcal{H}_\infty$ suboptimal controllers, we shall make a change of variables as in equation (16.4):

$$v = u + B_2^* X_\infty x, \quad r = w - \gamma^{-2} B_1^* X_\infty x.$$

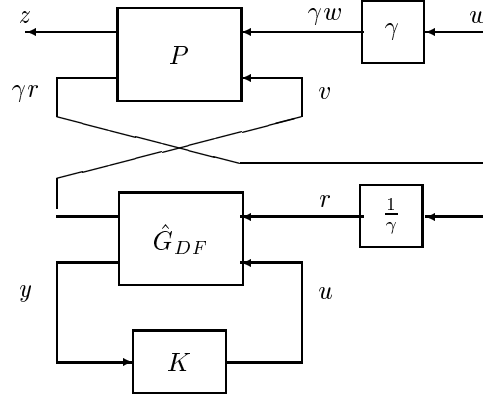
Then the system equations can be written as follows:

$$\begin{bmatrix} \dot{x} \\ z \\ \gamma r \end{bmatrix} = \begin{bmatrix} A_{F_\infty} & \gamma^{-1} B_1 & B_2 \\ C_{1F_\infty} & 0 & D_{12} \\ -\gamma^{-1} B_1^* X_\infty & I & 0 \end{bmatrix} \begin{bmatrix} x \\ \gamma w \\ v \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ v \\ y \end{bmatrix} = \begin{bmatrix} A_{tmp} & B_1 & B_2 \\ -F_\infty & 0 & I \\ C_2 + \gamma^{-2} B_1^* X_\infty & I & 0 \end{bmatrix} \begin{bmatrix} x \\ r \\ u \end{bmatrix}.$$

This is shown pictorially in the following diagram:



where

$$P = \left[\begin{array}{c|cc} A_{F_\infty} & \gamma^{-1}B_1 & B_2 \\ \hline C_{1F_\infty} & 0 & D_{12} \\ -\gamma^{-1}B_1^*X_\infty & I & 0 \end{array} \right]$$

and

$$\hat{G}_{DF} = \left[\begin{array}{c|cc} A_{tmp} & B_1 & B_2 \\ \hline -F_\infty & 0 & I \\ C_2 + \gamma^{-2}B_1^*X_\infty & I & 0 \end{array} \right].$$

Since $A_{tmp} - B_1(C_2 + \gamma^{-2}B_1^*X_\infty) = A - B_1C_2$ and $A_{tmp} + B_2F_\infty$ are stable, the rank conditions of Lemma 16.1 for system $\hat{G}_{DF}$ are satisfied. So from Theorem 16.2 and Lemma 16.7, we conclude that K is an admissible controller for G and $\|T_{zw}\|_\infty < \gamma$ iff K is an admissible controller for $\hat{G}_{DF}$ and $\|T_{vr}\|_\infty < \gamma$. Now it is easy to see by comparing this formula with the controller parameterization in Theorem 12.8 that $\mathcal{F}_\ell(M_\infty, Q)$ with $Q \in \mathcal{RH}_\infty$ (no norm constraint) parameterizes all stabilizing controllers for $\hat{G}_{DF}$; however, simple algebra shows that $T_{vr} = \mathcal{F}_\ell(\hat{G}_{DF}, \mathcal{F}_\ell(M_\infty, Q)) = Q$. So $\|T_{vr}\|_\infty < \gamma$ iff $\|Q\|_\infty < \gamma$, and $\mathcal{F}_\ell(M_\infty, Q)$ with $Q \in \mathcal{RH}_\infty$ and $\|Q\|_\infty < \gamma$ parameterizes all suboptimal controllers for G . $\square$

16.7 Output Estimation

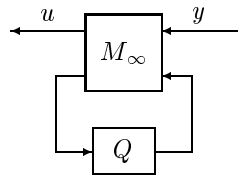
$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & I \\ C_2 & D_{21} & 0 \end{array} \right]$$

This problem is dual to DF, just as FC was to FI. Thus the discussion of the DF problem is relevant here, when appropriately dualized. The OE assumptions are

- (i) (A, B_1) is stabilizable and $A - B_2C_1$ is stable;
- (ii) (C_2, A) is detectable;
- (iv) $\begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^* = \begin{bmatrix} 0 \\ I \end{bmatrix}$.

Assumption (i), together with (iv), imply that internal stability is again equivalent to $T_{zw} \in \mathcal{RH}_\infty$, as in the output feedback case.

Theorem 16.13 *There exists an admissible controller for the OE problem such that $\|T_{zw}\|_\infty < \gamma$ if and only if $J_\infty \in \text{dom}(\text{Ric})$ and $\text{Ric}(J_\infty) \geq 0$. Moreover, if these conditions are satisfied then all admissible controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as the set of all transfer matrices from y to u in*



$$M_\infty(s) = \left[\begin{array}{c|cc} A + L_\infty C_2 - B_2 C_1 & L_\infty & -B_2 - \gamma^{-2} Y_\infty C_1^* \\ \hline C_1 & 0 & I \\ C_2 & I & 0 \end{array} \right]$$

with $Q \in \mathcal{RH}_\infty$, $\|Q\|_\infty < \gamma$.

It is interesting to compare $\mathcal{H}_\infty$ and $\mathcal{H}_2$ in the context of the OE problem, even though, by duality, the essence of these remarks was made before. Both optimal estimators are observers with the observer gain determined by $\text{Ric}(J_\infty)$ and $\text{Ric}(J_2)$. Optimal $\mathcal{H}_2$ output estimation consists of multiplying the optimal state estimate by the output map C_1 . Thus optimal $\mathcal{H}_2$ estimation depends only trivially on the output z that is being estimated, and *state* estimation is the fundamental problem. In contrast, the $\mathcal{H}_\infty$ estimation problem depends very explicitly and importantly on the output being estimated. This will have implications for the separation properties of the $\mathcal{H}_\infty$ output feedback controller.

16.8 Separation Theory

If we assume the results of the special problems, which are proven in the previous sections, we can now prove Theorems 16.4 and 16.5 using separation arguments. This essentially involves reducing the output feedback problem to a combination of the Full Information and the Output Estimation problems. The separation properties of the $\mathcal{H}_\infty$ controller are more complicated than the $\mathcal{H}_2$ controller, although they are no less interesting. The notation and assumptions for this section are as in Section 16.2.

16.8.1 $\mathcal{H}_\infty$ Controller Structure

The $\mathcal{H}_\infty$ controller formulae from Theorem 16.4 are

$$K_{sub}(s) := \left[\begin{array}{c|c} \hat{A}_\infty & -Z_\infty L_\infty \\ \hline F_\infty & 0 \end{array} \right]$$

$$\hat{A}_\infty := A + \gamma^{-2} B_1 B_1^* X_\infty + B_2 F_\infty + Z_\infty L_\infty C_2$$

$$F_\infty := -B_2^* X_\infty, \quad L_\infty := -Y_\infty C_2^*, \quad Z_\infty := (I - \gamma^{-2} Y_\infty X_\infty)^{-1}$$

where $X_\infty := Ric(H_\infty)$ and $Y_\infty := Ric(J_\infty)$. The necessary and sufficient conditions for the existence of an admissible controller such that $\|T_{zw}\|_\infty < \gamma$ are

- (i) $H_\infty \in dom(Ric)$ and $X_\infty := Ric(H_\infty) \geq 0$;
- (ii) $J_\infty \in dom(Ric)$ and $Y_\infty := Ric(J_\infty) \geq 0$;
- (iii) $\rho(X_\infty Y_\infty) < \gamma^2$.

We have seen that condition (i) corresponds to the Full Information condition and that (ii) corresponds to the Full Control condition. It is easily shown that, given the FI and FC results, these conditions are necessary for the output feedback case as well.

Lemma 16.14 *Suppose there exists an admissible controller making $\|T_{zw}\|_\infty < \gamma$. Then conditions (i) and (ii) hold.*

Proof. Let K be an admissible controller for which $\|T_{zw}\|_\infty < \gamma$. The controller $K[C_2 \ D_{21}]$ solves the FI problem; hence from Theorem 16.9, $H_\infty \in dom(Ric)$ and $X_\infty := Ric(H_\infty) \geq 0$. Condition (ii) follows by the dual argument. $\square$

We would also expect some condition beyond these two, and that is provided by (iii), which is an elegant combination of elements from FI and FC. Note that all the conditions of Theorem 16.4 are symmetric in H_∞ , J_∞ , X_∞ , and Y_∞ , but the formula for the controller is not. Needless to say, there is a dual form that can be obtained by inspection from the above formula. For a symmetric formula, the state equations above can be multiplied through by Z_∞^{-1} and put in descriptor form. A simple substitution from the Riccati equation for X_∞ will then yield a symmetric, though more complicated, formula:

$$(I - \gamma^{-2} Y_\infty X_\infty) \dot{\hat{x}} = A_s \hat{x} - L_\infty y \quad (16.10)$$

$$u = F_\infty \hat{x} \quad (16.11)$$

where $A_s := A + B_2 F_\infty + L_\infty C_2 + \gamma^{-2} Y_\infty A^* X_\infty + \gamma^{-2} B_1 B_1^* X_\infty + \gamma^{-2} Y_\infty C_1^* C_1$.

To emphasize its relationship to the $\mathcal{H}_2$ controller formulae, the $\mathcal{H}_\infty$ controller can be written as

$$\dot{\hat{x}} = A \hat{x} + B_1 \hat{w}_{worst} + B_2 u + Z_\infty L_\infty (C_2 \hat{x} - y)$$

$$u = F_\infty \hat{x}, \quad \hat{w}_{worst} = \gamma^{-2} B_1^* X_\infty \hat{x}.$$

These equations have the structure of an observer-based compensator. The obvious questions that arise when these formulae are compared with the $\mathcal{H}_2$ formulae are

- 1) Where does the term $B_1 \hat{w}_{worst}$ come from?
- 2) Why $Z_\infty L_\infty$ instead of L_∞ ?
- 3) Is there a separation interpretation of these formulae analogous to that for $\mathcal{H}_2$?

The proof of Theorem 16.4 reveals that there is a very well-defined separation interpretation of these formulae and that $w_{worst} := \gamma^{-2} B_1^* X_\infty x$ is, in some sense, a worst-case input for the Full Information problem. Furthermore, $Z_\infty L_\infty$ is actually the optimal filter gain for estimating $F_\infty x$, which is the optimal Full Information control input, in the presence of this worst-case input. It is therefore not surprising that $Z_\infty L_\infty$ should enter in the controller equations instead of L_∞ . The term $\hat{w}_{worst}$ may be thought of loosely as an estimate for w_{worst} .

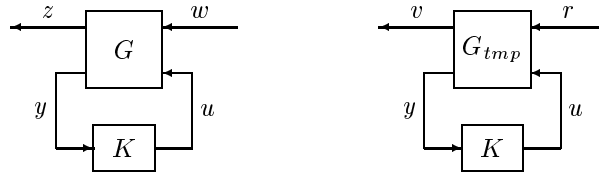
16.8.2 Proof of Theorem 16.4

It has been shown from Lemma 16.14 that conditions (i) and (ii) are necessary for $\|T_{zw}\|_\infty < \gamma$. Hence we only need to show that if conditions (i) and (ii) are satisfied, condition (iii) is necessary and sufficient for $\|T_{zw}\|_\infty < \gamma$. As in section 16.3, we define new disturbance and control variables

$$r := w - \gamma^{-2} B_1^* X_\infty x, \quad v := u + B_2^* X_\infty x.$$

Then

$$\begin{bmatrix} v \\ y \end{bmatrix} = \left[\begin{array}{c|cc} A_{tmp} & B_1 & B_2 \\ \hline -F_\infty & 0 & I \\ C_2 & D_{21} & 0 \end{array} \right] \begin{bmatrix} r \\ u \end{bmatrix} = G_{tmp} \begin{bmatrix} r \\ u \end{bmatrix} \quad A_{tmp} := A + \gamma^{-2} B_1 B_1^* X_\infty.$$



Recall from Lemma 16.8 that K is admissible for G and $\|T_{zw}\|_\infty < \gamma$ iff K is admissible for G_{tmp} and $\|T_{vr}\|_\infty < \gamma$.

While G_{tmp} has the form required for the OE problem, to actually use the OE results, we will need to verify that G_{tmp} satisfies the following assumptions for the OE problem:

- (i) (A_{tmp}, B_1) is stabilizable and $A_{tmp} + B_2 F_\infty$ is stable;
- (ii) (C_2, A_{tmp}) is detectable;
- (iv) $\begin{bmatrix} B_1 \\ D_{21} \end{bmatrix} D_{21}^* = \begin{bmatrix} 0 \\ I \end{bmatrix}$.

Assumption (iv) and that (A_{tmp}, B_1) is stabilizable follow immediately from the corresponding assumptions for Theorem 16.4. The stability of $A_{tmp} + B_2 F_\infty$ follows from the definition of $H_\infty \in \text{dom}(\text{Ric})$. The following lemma gives conditions for assumption (ii) to hold. Of course, the existence of an admissible controller for G_{tmp} immediately implies that assumption (ii) holds. Note that the OE Hamiltonian matrix for G_{tmp} is

$$J_{tmp} := \begin{bmatrix} A_{tmp}^* & \gamma^{-2} F_\infty^* F_\infty - C_2^* C_2 \\ -B_1 B_1^* & -A_{tmp} \end{bmatrix}.$$

Lemma 16.15 *If $J_{tmp} \in \text{dom}(\text{Ric})$ and $Y_{tmp} := \text{Ric}(J_{tmp}) \geq 0$, then (C_2, A_{tmp}) is detectable.*

Proof. The lemma follows from the dual to Lemma 16.6, which gives that $(A_{tmp} - Y_{tmp} C_2^* C_2)$ is stable. $\square$

Proof of Theorem 16.4 (Sufficiency) Assume the conditions (i) through (iii) in the theorem statement hold. Using the Riccati equation for X_∞ , one can easily verify that

$T := \begin{bmatrix} I & -\gamma^{-2} X_\infty \\ 0 & I \end{bmatrix}$ provides a similarity transformation between J_{tmp} and J_∞ , i.e., $T^{-1} J_{tmp} T = J_\infty$. So

$$\mathcal{X}_-(J_{tmp}) = T \mathcal{X}_-(J_\infty) = T \text{Im} \begin{bmatrix} I \\ Y_\infty \end{bmatrix} = \text{Im} \begin{bmatrix} I - \gamma^{-2} X_\infty Y_\infty \\ Y_\infty \end{bmatrix}$$

and $\rho(X_\infty Y_\infty) < \gamma^2$ implies that $J_{tmp} \in \text{dom}(\text{Ric})$ and $Y_{tmp} := \text{Ric}(J_{tmp}) = Y_\infty (I - \gamma^{-2} X_\infty Y_\infty)^{-1} = Z_\infty Y_\infty \geq 0$. Thus by Lemma 16.15 the OE assumptions hold for G_{tmp} , and by Theorem 16.13 the OE problem is solvable. From Theorem 16.13 with $Q = 0$, one solution is

$$\left[\begin{array}{c|c} \frac{A + \gamma^{-2} B_1 B_1^* X_\infty - Y_{tmp} C_2^* C_2 + B_2 F_\infty}{F_\infty} & Y_{tmp} C_2^* \\ \hline & 0 \end{array} \right]$$

but this is precisely K_{sub} defined in Theorem 16.4. We conclude that K_{sub} stabilizes G_{tmp} and that $\|T_{vr}\|_\infty < \gamma$. Then by Lemma 16.8, K_{sub} stabilizes G and that $\|T_{zw}\|_\infty < \gamma$.

(Necessity) Let K be an admissible controller for which $\|T_{zw}\|_\infty < \gamma$. By Lemma 16.14, $H_\infty \in \text{dom}(\text{Ric})$, $X_\infty := \text{Ric}(H_\infty) \geq 0$, $J_\infty \in \text{dom}(\text{Ric})$, and $Y_\infty := \text{Ric}(J_\infty) \geq 0$. From Lemma 16.8, K is admissible for G_{tmp} and $\|T_{vr}\|_\infty < \gamma$. This implies that the OE assumptions hold for G_{tmp} and that the OE problem is solvable. Therefore, from Theorem 16.13 applied to G_{tmp} , we have that $J_{tmp} \in \text{dom}(\text{Ric})$ and $Y_{tmp} = \text{Ric}(J_{tmp}) \geq 0$. Using the same similarity transformation formula as in the sufficiency part, we get that $Y_{tmp} = (I - \gamma^{-2}Y_\infty X_\infty)^{-1}Y_\infty \geq 0$. We shall now show that $Y_{tmp} \geq 0$ implies that $\rho(X_\infty Y_\infty) < \gamma^2$. We shall consider two cases:

- Y_∞ is nonsingular: in this case $Y_{tmp} \geq 0$ implies that $I - \gamma^{-2}Y_\infty^{1/2}X_\infty Y_\infty^{1/2} > 0$. So $\rho(Y_\infty^{1/2}X_\infty Y_\infty^{1/2}) < \gamma^2$ or $\rho(X_\infty Y_\infty) < \gamma^2$.
- Y_∞ is singular: there is a unitary matrix U such that

$$Y_\infty = U^* \begin{bmatrix} Y_{11} & 0 \\ 0 & 0 \end{bmatrix} U$$

with $Y_{11} > 0$. Let $UX_\infty U^*$ be partitioned accordingly,

$$UX_\infty U^* = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix}.$$

Then by the same argument as in the Y_∞ nonsingular case,

$$Y_{tmp} = U^* \begin{bmatrix} (I - \gamma^{-2}Y_{11}X_{11})^{-1}Y_{11} & 0 \\ 0 & I \end{bmatrix} U \geq 0$$

implies that $\gamma^2 > \rho(X_{11}Y_{11}) (= \rho(X_\infty Y_\infty))$. $\square$

We now see exactly why the term involving $\hat{w}_{worst}$ appears and why the “observer” gain is $Z_\infty L_\infty$. Both terms are consequences of estimating the optimal Full Information (i.e., state feedback) control gain. While an analogous output estimation problem arises in the $\mathcal{H}_2$ output feedback problem, the resulting equations are much simpler. This is because there is no “worst-case” disturbance for the $\mathcal{H}_2$ Full Information problem and because the problem of estimating any output, including the optimal state feedback, is equivalent to state estimation.

We now present a separation interpretation for $\mathcal{H}_\infty$ suboptimal controllers. It will be stated in terms of the central controller, but similar interpretations could be made for the parameterization of all suboptimal controllers (see the proofs of Theorems 16.4 and 16.5).

The $\mathcal{H}_\infty$ output feedback controller is the output estimator of the full information control law in the presence of the “worst-case” disturbance w_{worst} .

Note that the same statement holds for the $\mathcal{H}_2$ optimal controller, except that $w_{worst} = 0$.

16.8.3 Proof of Theorem 16.5

From Lemma 16.8, the set of all admissible controllers for G such that $\|T_{zw}\|_\infty < \gamma$ equals the set of all admissible controllers for G_{tmp} such that $\|T_{vr}\|_\infty < \gamma$. Apply Theorem 16.13. $\square$

16.9 Optimality and Limiting Behavior

In this section, we will discuss, without proof, the behavior of the $\mathcal{H}_\infty$ suboptimal solution as γ varies, especially as γ approaches the infimal achievable norm, denoted by γ_{opt} . Since Theorem 16.4 gives necessary and sufficient conditions for the existence of an admissible controller such that $\|T_{zw}\|_\infty < \gamma$, γ_{opt} is the infimum over all γ such that conditions (i)-(iii) are satisfied. Theorem 16.4 does not give an explicit formula for γ_{opt} , but, just as for the $\mathcal{H}_\infty$ norm calculation, it can be computed as closely as desired by a search technique.

Although we have not focused on the problem of $\mathcal{H}_\infty$ optimal controllers, the assumptions in this book make them relatively easy to obtain in most cases. In addition to describing the qualitative behavior of suboptimal solutions as γ varies, we will indicate why the descriptor version of the controller formulae from Section 16.8.1 can usually provide formulae for the optimal controller when $\gamma = \gamma_{opt}$. Most of these results can be obtained relatively easily using the machinery that is developed in the previous sections. The reader interested in filling in the details is encouraged to begin by strengthening assumption (i) to controllable and observable and considering the Hamiltonians for X_∞^{-1} and Y_∞^{-1} .

As $\gamma \rightarrow \infty$, $H_\infty \rightarrow H_2$, $X_\infty \rightarrow X_2$, etc., and $K_{sub} \rightarrow K_2$. This fact is the result of the particular choice for the central controller ($Q = 0$) that was made here. While it could be argued that K_{sub} is a natural choice, this connection with $\mathcal{H}_2$ actually hints at deeper interpretations. In fact, K_{sub} is the minimum entropy solution (see next section) as well as the minimax controller for $\|z\|_2^2 - \gamma^2 \|w\|_2^2$.

If $\gamma_2 \geq \gamma_1 > \gamma_{opt}$, then $X_\infty(\gamma_1) \geq X_\infty(\gamma_2)$ and $Y_\infty(\gamma_1) \geq Y_\infty(\gamma_2)$. Thus X_∞ and Y_∞ are decreasing functions of γ , as is $\rho(X_\infty Y_\infty)$. At $\gamma = \gamma_{opt}$, anyone of the three conditions in Theorem 16.4 can fail. If only condition (iii) fails, then it is relatively straightforward to show that the descriptor formulae for $\gamma = \gamma_{opt}$ are optimal, i.e., the optimal controller is given by

$$(I - \gamma_{opt}^{-2} Y_\infty X_\infty) \dot{\hat{x}} = A_s \hat{x} - L_\infty y \quad (16.12)$$

$$u = F_\infty \hat{x} \quad (16.13)$$

where $A_s := A + B_2 F_\infty + L_\infty C_2 + \gamma_{opt}^{-2} Y_\infty A^* X_\infty + \gamma_{opt}^{-2} B_1 B_1^* X_\infty + \gamma_{opt}^{-2} Y_\infty C_1^* C_1$. See the example below.

The formulae in Theorem 16.4 are not well-defined in the optimal case because the term $(I - \gamma_{opt}^{-2} X_\infty Y_\infty)$ is not invertible. It is possible but far less likely that conditions (i) or (ii) would fail before (iii). To see this, consider (i) and let γ_1 be the largest γ

for which H_∞ fails to be in $\text{dom}(\text{Ric})$ because the H_∞ matrix fails to have either the stability property or the complementarity property. The same remarks will apply to (ii) by duality.

If complementarity fails at $\gamma = \gamma_1$, then $\rho(X_\infty) \rightarrow \infty$ as $\gamma \rightarrow \gamma_1$. For $\gamma < \gamma_1$, H_∞ may again be in $\text{dom}(\text{Ric})$, but X_∞ will be indefinite. For such γ , the controller $u = -B_2^* X_\infty x$ would make $\|T_{zw}\|_\infty < \gamma$ but would not be stabilizing. See part 1) of the example below. If the stability property fails at $\gamma = \gamma_1$, then $H_\infty \notin \text{dom}(\text{Ric})$ but Ric can be extended to obtain X_∞ and the controller $u = -B_2^* X_\infty x$ is stabilizing and makes $\|T_{zw}\|_\infty = \gamma_1$. The stability property will also not hold for any $\gamma \leq \gamma_1$, and no controller whatsoever exists which makes $\|T_{zw}\|_\infty < \gamma_1$. In other words, if stability breaks down first, then the infimum over stabilizing controllers equals the infimum over all controllers, stabilizing or otherwise. See part 2) of the example below. In view of this, we would typically expect that complementarity would fail first.

Complementarity failing at $\gamma = \gamma_1$ means $\rho(X_\infty) \rightarrow \infty$ as $\gamma \rightarrow \gamma_1$, so condition (iii) would fail at even larger values of γ , unless the eigenvectors associated with $\rho(X_\infty)$ as $\gamma \rightarrow \gamma_1$ are in the null space of Y_∞ . Thus condition (iii) is the most likely of all to fail first. If condition (i) or (ii) fails first because the stability property fails, the formulae in Theorem 16.4 as well as their descriptor versions are optimal at $\gamma = \gamma_{opt}$. This is illustrated in the example below for the output feedback. If the complementarity condition fails first, (but (iii) does not fail), then obtaining formulae for the optimal controllers is a more subtle problem.

Example 16.1 Let an interconnected dynamical system realization be given by

$$G(s) = \left[\begin{array}{c|cc} a & \begin{bmatrix} 1 & 0 \end{bmatrix} & b_2 \\ \hline \begin{bmatrix} 1 \\ 0 \end{bmatrix} & 0 & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \hline c_2 & \begin{bmatrix} 0 & 1 \end{bmatrix} & 0 \end{array} \right]$$

with $|c_2| \geq |b_2| > 0$. Then all assumptions for the output feedback problem are satisfied and

$$H_\infty = \begin{bmatrix} a & \frac{1-b_2^2\gamma^2}{\gamma^2} \\ -1 & -a \end{bmatrix}, \quad J_\infty = \begin{bmatrix} a & \frac{1-c_2^2\gamma^2}{\gamma^2} \\ -1 & -a \end{bmatrix}.$$

The eigenvalues of H_∞ and J_∞ are given, respectively, by

$$\sigma(H_\infty) = \left\{ \pm \frac{\sqrt{(a^2 + b_2^2)\gamma^2 - 1}}{\gamma} \right\}, \quad \sigma(J_\infty) = \left\{ \pm \frac{\sqrt{(a^2 + c_2^2)\gamma^2 - 1}}{\gamma} \right\}.$$

If $\gamma^2 > \frac{1}{a^2 + b_2^2} \left(\geq \frac{1}{a^2 + c_2^2} \right)$, then $\mathcal{X}_-(H_\infty)$ and $\mathcal{X}_-(J_\infty)$ exist and

$$\mathcal{X}_-(H_\infty) = \text{Im} \begin{bmatrix} \frac{\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma}{\gamma} \\ 1 \end{bmatrix}$$

$$\mathcal{X}_-(J_\infty) = \text{Im} \begin{bmatrix} \frac{\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma}{\gamma} \\ 1 \end{bmatrix}.$$

We shall consider two cases:

- 1) $a > 0$: In this case, the complementary property of $\text{dom}(\text{Ric})$ will fail before the stability property fails since

$$\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma = 0$$

$$\text{when } \gamma^2 = \frac{1}{b_2^2} \quad \left(> \frac{1}{a^2 + b_2^2} \right).$$

Nevertheless, if $\gamma^2 > \frac{1}{a^2 + b_2^2}$ and $\gamma^2 \neq \frac{1}{b_2^2}$, then $H_\infty \in \text{dom}(\text{Ric})$ and

$$X_\infty = \frac{\gamma}{\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma} = \begin{cases} > 0; & \text{if } \gamma^2 > \frac{1}{b_2^2} \\ < 0; & \text{if } \frac{1}{a^2 + b_2^2} < \gamma^2 < \frac{1}{b_2^2}. \end{cases}$$

Let $F_\infty = -B_2^* X_\infty$; then

$$A + B_2 F_\infty = -\frac{a + b_2^2 \gamma \sqrt{(a^2 + b_2^2)\gamma^2 - 1}}{b_2^2 \gamma^2 - 1} = \begin{cases} < 0 \text{ (stable)}; & \text{if } \gamma^2 > \frac{1}{b_2^2} \\ > 0 \text{ (unstable)}; & \text{if } \frac{1}{a^2 + b_2^2} < \gamma^2 < \frac{1}{b_2^2}. \end{cases}$$

– Suppose full information (or states) are available for feedback and let

$$u = F_\infty x.$$

Then the closed-loop transfer matrix is given by

$$T_{zw} = \left[\begin{array}{c|c} A + B_2 F_\infty & B_1 \\ \hline C_1 + D_{12} F_\infty & 0 \end{array} \right] = \left[\begin{array}{c|c} \frac{-a + b_2^2 \gamma \sqrt{(a^2 + b_2^2)\gamma^2 - 1}}{b_2^2 \gamma^2 - 1} & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 1 \\ \frac{-b_2 \gamma}{\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma} \end{bmatrix} & 0 \end{array} \right],$$

and T_{zw} is stable for all $\gamma^2 > \frac{1}{b_2^2}$ and is not stable for $\frac{1}{a^2 + b_2^2} < \gamma^2 < \frac{1}{b_2^2}$. Furthermore, it can be shown that $\|T_{zw}\| < \gamma$ for all $\gamma^2 > \frac{1}{a^2 + b_2^2}$ and $\gamma^2 \neq \frac{1}{b_2^2}$.

It is clear that the optimal $\mathcal{H}_\infty$ norm is $\frac{1}{b_2}$ but is not achievable.

- Suppose the states are not available; then output feedback must be considered. Note that if $\gamma^2 > \frac{1}{b_2^2}$, then $H_\infty \in \text{dom}(\text{Ric})$, $J_\infty \in \text{dom}(\text{Ric})$, and

$$X_\infty = \frac{\gamma}{\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma} > 0$$

$$Y_\infty = \frac{\gamma}{\sqrt{(a^2 + c_2^2)\gamma^2 - 1} - a\gamma} > 0.$$

Hence conditions (i) and (ii) in Theorem 16.4 are satisfied, and need to check condition (iii). Since

$$\rho(X_\infty Y_\infty) = \frac{\gamma^2}{(\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma)(\sqrt{(a^2 + c_2^2)\gamma^2 - 1} - a\gamma)},$$

it is clear that $\rho(X_\infty Y_\infty) \rightarrow \infty$ when $\gamma^2 \rightarrow \frac{1}{b_2^2}$. So condition (iii) will fail before condition (i) or (ii) fails.

- 2) $a < 0$: In this case, complementary property is always satisfied, and, furthermore, $H_\infty \in \text{dom}(\text{Ric})$, $J_\infty \in \text{dom}(\text{Ric})$, and

$$X_\infty = \frac{\gamma}{\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma} > 0$$

$$Y_\infty = \frac{\gamma}{\sqrt{(a^2 + c_2^2)\gamma^2 - 1} - a\gamma} > 0$$

for $\gamma^2 > \frac{1}{a^2 + b_2^2}$.

However, for $\gamma^2 \leq \frac{1}{a^2 + b_2^2}$, $H_\infty \notin \text{dom}(\text{Ric})$ since stability property fails. Nevertheless, in this case, if $\gamma_0^2 = \frac{1}{a^2 + b_2^2}$, we can extend the $\text{dom}(\text{Ric})$ to include those matrices H_∞ with imaginary axis eigenvalues as

$$\overline{\mathcal{X}}_-(H_\infty) = \text{Im} \begin{bmatrix} -a \\ 1 \end{bmatrix}$$

such that $X_\infty = -\frac{1}{a}$ is a solution to the Riccati equation

$$A^* X_\infty + X_\infty A + C_1^* C_1 + \gamma_0^{-2} X_\infty B_1 B_1^* X_\infty - X_\infty B_2 B_2^* X_\infty = 0$$

and $A + \gamma_0^{-2} B_1 B_1^* X_\infty - B_2 B_2^* X_\infty = 0$.

- For $\gamma = \gamma_0$ and $F_\infty = -B_2^* X_\infty$, then $A + B_2 F_\infty = a + \frac{b_2^2}{a} < 0$. So if states are available for feedback and $u = F_\infty x$, we have

$$T_{zw} = \left[\begin{array}{c|c} a + \frac{b_2^2}{a} & \begin{bmatrix} 1 & 0 \end{bmatrix} \\ \hline \begin{bmatrix} 1 \\ \frac{b_2}{a} \end{bmatrix} & 0 \end{array} \right] \in \mathcal{RH}_\infty$$

and $\|T_{zw}\|_\infty = \frac{1}{\sqrt{a^2 + b_2^2}} = \gamma_0$. Hence the optimum is achieved.

- If states are not available, the output feedback is considered, and $|b_2| = |c_2|$, then it can be shown that

$$\rho(X_\infty Y_\infty) = \frac{\gamma^2}{(\sqrt{(a^2 + b_2^2)\gamma^2 - 1} - a\gamma)^2} < \gamma^2$$

is satisfied if and only if

$$\gamma > \frac{\sqrt{a^2 + 2b_2^2} + a}{b_2^2} \left(> \frac{1}{\sqrt{a^2 + b_2^2}} \right).$$

So condition (iii) of Theorem 16.4 will fail before either (i) or (ii) fails.

In both $a > 0$ and $a < 0$ cases, the optimal γ for the output feedback is given by

$$\gamma_{opt} = \frac{\sqrt{a^2 + 2b_2^2} + a}{b_2^2}$$

if $|b_2| = |c_2|$; and the optimal controller given by the descriptor formula in equations (16.12) and (16.13) is a constant. In fact,

$$u_{opt} = -\frac{\gamma_{opt}}{\sqrt{(a^2 + b_2^2)\gamma_{opt}^2 - 1} - a\gamma_{opt}} y.$$

For instance, let $a = -1$ and $b_2 = 1 = c_2$. Then $\gamma_{opt} = \sqrt{3} - 1 = 0.7321$ and $u_{opt} = -0.7321 y$. Further,

$$T_{zw} = \left[\begin{array}{c|cc} -1.7321 & 1 & -0.7321 \\ \hline 1 & 0 & 0 \\ -0.7321 & 0 & -0.7321 \end{array} \right].$$

It is easy to check that $\|T_{zw}\|_\infty = 0.7321$. ◇

16.10 Controller Interpretations

This section considers some additional connections with the minimum entropy solution and the work of Whittle and will be of interest primarily to readers already familiar with them. The connection with the Q -parameterization approach will be considered in the next chapter for the general case.

Section 16.10.2 gives another separation interpretation of the central $\mathcal{H}_\infty$ controller of Theorem 16.4 in the spirit of Whittle (1981). It has been shown in Glover and Doyle [1988] that the central controller corresponds exactly to the steady state version of the optimal risk sensitive controller derived by [Whittle, 1981], who also derives a separation result and a certainty equivalence principle (see also [Whittle, 1986]).

16.10.1 Minimum Entropy Controller

Let T be a transfer matrix with $\|T\|_\infty < \gamma$. Then the entropy of $T(s)$ is defined by

$$I(T, \gamma) = -\frac{\gamma^2}{2\pi} \int_{-\infty}^{\infty} \ln |\det (I - \gamma^{-2} T^*(j\omega) T(j\omega))| d\omega.$$

It is easy to see that

$$I(T, \gamma) = -\frac{\gamma^2}{2\pi} \int_{-\infty}^{\infty} \sum_i \ln |1 - \gamma^{-2} \sigma_i^2(T(j\omega))| d\omega$$

and $I(T, \gamma) \geq 0$, where $\sigma_i(T(j\omega))$ is the i th singular value of $T(j\omega)$. It is also easy to show that

$$\lim_{\gamma \rightarrow \infty} I(T, \gamma) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \sum_i \sigma_i^2(T(j\omega)) d\omega = \|T\|_2^2.$$

Thus the entropy $I(T, \gamma)$ is in fact a performance index measuring the tradeoff between the $\mathcal{H}_\infty$ optimality ($\gamma \rightarrow \|T\|_\infty$) and the $\mathcal{H}_2$ optimality ($\gamma \rightarrow \infty$).

It has been shown in Glover and Mustafa [1989] that the central controller given in Theorem 16.4 is actually the controller that satisfies the norm condition $\|T_{zw}\|_\infty < \gamma$ and minimizes the following entropy:

$$-\frac{\gamma^2}{2\pi} \int_{-\infty}^{\infty} \ln |\det (I - \gamma^{-2} T_{zw}^*(j\omega) T_{zw}(j\omega))| d\omega.$$

Therefore, the central controller is also called the minimum entropy controller (maximum entropy controller if the entropy is defined as $\tilde{I}(T, \gamma) = -I(T, \gamma)$).

16.10.2 Relations with Separation in Risk Sensitive Control

Although [Whittle, 1981] treats a finite horizon, discrete time, stochastic control problem, his separation result has a clear interpretation for the present infinite horizon,

continuous time, deterministic control problem, as given below; and it is an interesting exercise to compare the two separation statements. This discussion will be entirely in the time-domain.

We will consider the system at time, $t = 0$, and evaluate the *past stress*, $\mathcal{S}_-$, and *future stress*, $\mathcal{S}_+$, as functions of the current state, x . First define the future stress as

$$\mathcal{S}_+(x) := \sup_w \inf_u (\|P_+ z\|_2^2 - \gamma^2 \|P_+ w\|_2^2);$$

then by the completion of the squares and by the saddle point argument of Section 16.3, where u is not constrained to be a function of the measurements (FI case), we obtain

$$\mathcal{S}_+(x) = x^* X_\infty x.$$

The *past stress*, $\mathcal{S}_-(x)$, is a function of the past inputs and observations, $u(t), y(t)$ for $-\infty < t < 0$, and of the present state, x , and is produced by the worst case disturbance, w , that is consistent with the given data:

$$\mathcal{S}_-(x) := \sup (\|P_- z\|_2^2 - \gamma^2 \|P_- w\|_2^2).$$

In order to evaluate $\mathcal{S}_-$ we see that w can be divided into two components, $D_{21}w$ and $D_{21}^\perp w$, with x dependent only on $D_{21}^\perp w$ (since $B_1 D_{21}^* = 0$) and $D_{21}w = y - C_2 x$. The past stress is then calculated by a completion of the square and in terms of a filter output. In particular, let $\bar{x}$ be given by the stable differential equation

$$\dot{\bar{x}} = A\bar{x} + B_2 u + L_\infty (C_2 \bar{x} - y) + Y_\infty C_1^* C_1 \bar{x} \quad \text{with } \bar{x}(-\infty) = 0.$$

Then it can be shown that the worst case w is given by

$$D_{21}^\perp w = D_{21}^\perp B_1^* Y_\infty^{-1} (x(t) - \bar{x}(t)) \quad \text{for } t < 0$$

and that this gives, with $e := x - \bar{x}$,

$$\mathcal{S}_-(x) = -\gamma^2 e(0)^* Y_\infty^{-1} e(0) - \gamma^2 \|P_-(y - C_2 \bar{x})\|_2^2 + \|P_-(C_1 \bar{x})\|_2^2 + \|P_- u\|_2^2.$$

The worst case disturbance will now reach the value of x to maximize the total stress, $\mathcal{S}_-(x) + \mathcal{S}_+(x)$, and this is easily shown to be achieved at the current state of

$$\hat{x} = Z_\infty \bar{x}(0).$$

The definitions of X_∞ and Y_∞ can be used to show that the state equations for the central controller can be rewritten with the state $\bar{x} := Z_\infty \hat{x}$ and with $\bar{x}$ as defined above. The control signal is then

$$u = F_\infty \hat{x} = F_\infty Z_\infty \bar{x}.$$

The *separation* is between the evaluation of future stress, which is a control problem with an unconstrained input, and the past stress, which is a filtering problem with known control input. The central controller then combines these evaluations to give a worst case estimate, $\hat{x}$, and the control law acts as if this were the perfectly observed state.

16.11 An Optimal Controller

To offer a general idea about the appearance of an optimal controller, we shall give in the following without proof the conditions under which an optimal controller exists and an explicit formula for an optimal controller.

Theorem 16.16 *There exists an admissible controller such that $\|T_{zw}\|_\infty \leq \gamma$ iff the following three conditions hold:*

(i) *there exists a full column rank matrix*

$$\begin{bmatrix} X_{\infty 1} \\ X_{\infty 2} \end{bmatrix} \in \mathbb{R}^{2n \times n}$$

such that

$$H_\infty \begin{bmatrix} X_{\infty 1} \\ X_{\infty 2} \end{bmatrix} = \begin{bmatrix} X_{\infty 1} \\ X_{\infty 2} \end{bmatrix} T_X, \quad \operatorname{Re} \lambda_i(T_X) \leq 0 \quad \forall i$$

and

$$X_{\infty 1}^* X_{\infty 2} = X_{\infty 2}^* X_{\infty 1};$$

(ii) *there exists a full column rank matrix*

$$\begin{bmatrix} Y_{\infty 1} \\ Y_{\infty 2} \end{bmatrix} \in \mathbb{R}^{2n \times n}$$

such that

$$J_\infty \begin{bmatrix} Y_{\infty 1} \\ Y_{\infty 2} \end{bmatrix} = \begin{bmatrix} Y_{\infty 1} \\ Y_{\infty 2} \end{bmatrix} T_Y, \quad \operatorname{Re} \lambda_i(T_Y) \leq 0 \quad \forall i$$

and

$$Y_{\infty 1}^* Y_{\infty 2} = Y_{\infty 2}^* Y_{\infty 1};$$

(iii)

$$\begin{bmatrix} X_{\infty 2}^* X_{\infty 1} & \gamma^{-1} X_{\infty 2}^* Y_{\infty 2} \\ \gamma^{-1} Y_{\infty 2}^* X_{\infty 2} & Y_{\infty 2}^* Y_{\infty 1} \end{bmatrix} \geq 0.$$

Moreover, when these conditions hold, one such controller is

$$K_{opt}(s) := C_K(sE_K - A_K)^+ B_K$$

where

$$\begin{aligned} E_K &:= Y_{\infty 1}^* X_{\infty 1} - \gamma^{-2} Y_{\infty 2}^* X_{\infty 2} \\ B_K &:= Y_{\infty 2}^* C_2^* \\ C_K &:= -B_2^* X_{\infty 2} \\ A_K &:= E_K T_X - B_K C_2 X_{\infty 1} = T_Y^* E_K + Y_{\infty 1}^* B_2 C_K. \end{aligned}$$

Remark 16.4 It is simple to show that if $X_{\infty 1}$ and $Y_{\infty 1}$ are nonsingular and if $X_{\infty} = X_{\infty 2} X_{\infty 1}^{-1}$ and $Y_{\infty} = Y_{\infty 2} Y_{\infty 1}^{-1}$, then condition (iii) in the above theorem is equivalent to $X_{\infty} \geq 0$, $Y_{\infty} \geq 0$, and $\rho(Y_{\infty} X_{\infty}) \leq \gamma^2$. So in this case, the conditions for the existence of an optimal controller can be obtained from “taking the limit” of the corresponding conditions in Theorem 16.4. Moreover, the controller given above is reduced to the descriptor form given in equations (16.12) and (16.13). $\heartsuit$

16.12 Notes and References

This chapter is based on Doyle, Glover, Khargonekar, Francis [1989], and Zhou [1992]. The minimum entropy controller is studied in detail in Glover and Mustafa [1989] and Mustafa and Glover [1990]. The risk sensitivity problem is treated in Whittle [1981, 1986]. The connections between the risk sensitivity controller and the central $\mathcal{H}_{\infty}$ controller are explored in Doyle and Glover [1988]. The complete characterization of optimal controllers for the general setup can be found in Glover, Limebeer, Doyle, Kasenally, and Safonov [1991].

